Own stock vs marketplace listing

v3.1 · Joybuy's UK and European business · Benchmark: Amazon UK published fees, a stand-in until Joybuy publishes its own.

This is a decision-support overview, written so colleagues can review the method, challenge the assumptions, and improve the model. It asks, for five UK categories, whether a marketplace should stock a representative line itself (1P) or leave it with a third-party seller (3P). It is not a forecast, not a profit claim, and not Joybuy's actual numbers. Joybuy has not published a seller fee card. Contribution is not profit. Prices and weekly demand are Assumptions, in pounds, exclusive of the UK's 20% VAT, and were not scraped from retailer listings.

Purpose

The central lens is a scale-up entrant (Joybuy-like) versus an established marketplace (Amazon UK-like), and the 1P versus 3P mix that follows. v3.1 is a principled fix, not a refit: v3.0 recommended 0% 1P at the entrant, while Joybuy launched ~100% 1P. The model was wrong, not Joybuy. JD China's 78.2% 1P revenue share is a parent-company figure, not Joybuy's mix, and is not a target. Scenario A treats 3P as available only where sellers can actually supply (default coverage 0%, so the real alternative is 'not ranged'); uses a 5% relative margin haircut (0–10% band; 25% is a stress test), justified by JD's published supplier scale; and charges only variable fulfilment to the unit, because already-built warehouse fixed cost is sunk. A land-grab layer, shown separately from unit economics, values range completeness. Scenario B uses high seller coverage, published retailer benchmarks, fully loaded fulfilment, Amazon UK published referral as a labelled stand-in, and a material ad take. A maturity slider moves the mix from 1P-led toward more 3P as seller depth and ads grow. All entrant parameters are Assumptions with stated reasoning.

The two scenarios, side by side

A · Entrant (Joybuy-like)

A scale-up marketplace, in the spirit of Joybuy's UK launch. Seller coverage is assumed at 0%: Joybuy launched 1P-only, so the real alternative to 1P is 'not ranged', not a 3P take. 1P gross margin is close to the Currys, Zalando and THG benchmarks (a 5% relative haircut), because Joybuy can draw on JD's published supplier and manufacturer relationships. Per-unit fulfilment uses variable cost only; already-built warehouse fixed costs are sunk and are shown as a scale-up investment, not a per-SKU penalty. Same-day subsidy is a customer-acquisition cost, shown separately. A land-grab layer, separate from unit economics, values range completeness, warehouse fill and the absence of 3P ads. Every number in this scenario is an Assumption with stated reasoning.

Operating (strategic) mid-band. Bring in-house 20 · Test 0 · Keep 3P 0 · Do not range 0. Implied 1P mix: 100% of sub-categories, 100% of modelled sales.
Unit-economics mid-band. Bring in-house 20 · Test 0 · Keep 3P 0 · Do not range 0. 1P mix: 100% of sub-categories, 100% of modelled sales (Assumption: ex-VAT price × assumed weekly units).

B · Established (Amazon UK-like)

A mature marketplace, in the spirit of Amazon UK. Seller coverage is assumed at 90%: a deep seller base can actually supply the range, so 3P is a real alternative. 1P margins and fulfilment use the published Currys, Zalando and THG benchmarks (fully loaded fulfilment at scale). Commission is the Amazon UK published referral schedule, labelled as a benchmark stand-in until Joybuy publishes its own. Advertising take is a 3% model estimate, informed by Amazon's published advertising revenue (US$68,635 million (FY2025)) but not computed as ads over 3P GMV, which Amazon does not publish. Amazon has disclosed that third-party sellers accounted for 61% of worldwide paid units (Q4 2025) (Q4 2025 earnings release, excluding Whole Foods). That is published context for an established *unit mix*, not a coverage ratio and not a target this model is fitted to. JD China's 78.2% 1P share is a *revenue* share (1P gross, 3P net) for the parent in China, and is not Joybuy's mix.

Operating mid-band (unit and strategic coincide). Bring in-house 5 · Test 3 · Keep 3P 12 · Do not range 0.
Implied 1P mix. 25% of sub-categories, 6% of modelled mid-band sales.

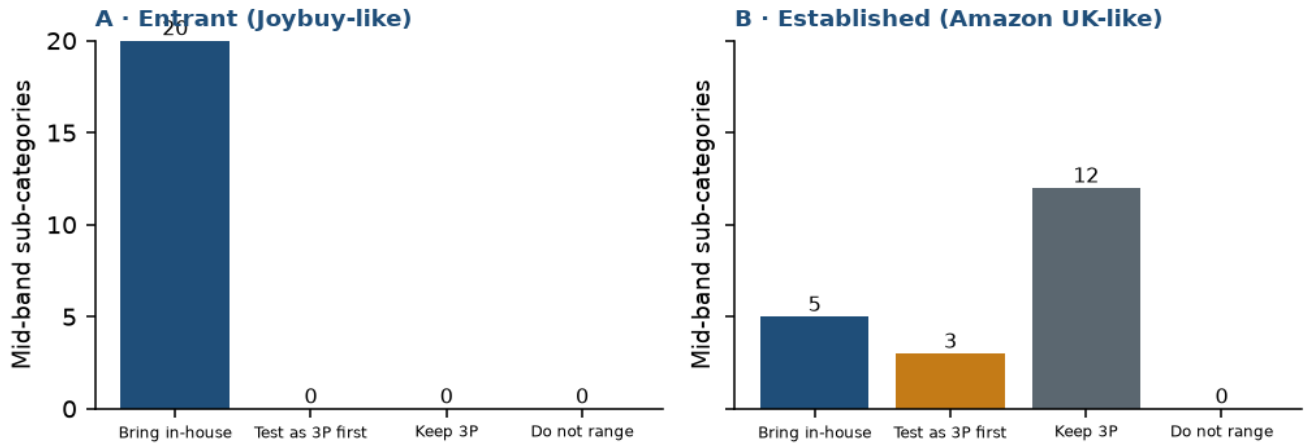


Figure 1. Mid-band recommendations under the default assumptions. Operating (strategic) recommendations. Bring in-house is the 1P path; Test as 3P first and Keep 3P are the 3P path; Do not range is a hole, not 3P. These are model estimates, not Joybuy decisions.

Implied 1P mix, mid-band. Entrant (A), operating / strategic: 100% of sub-categories and 100% of modelled sales. The same run's unit-economics mix is 100% of sub-categories and 100% of modelled sales. Established (B): 25% of sub-categories and 6% of modelled sales (unit and strategic coincide). 'Bring in-house' is the 1P path; 'Test as 3P first' and 'Keep 3P' are the 3P path; 'Do not range' is a hole, not 3P. Modelled sales are an Assumption.

Why an entrant tends to start 1P, then shift toward 3P

An entrant tends to start with first-party retail, and then lean more on third-party sellers as the platform matures. In v3.1 this is modelled in two layers, which are shown separately.

1. The real alternative is often 'not ranged', not 3P. At launch the seller base cannot actually supply the range. Comparing 1P contribution with a 3P take that is not on offer is the wrong comparison (that was the v3.0 mistake). The unit layer now weights the 3P payoff by seller coverage, and values the lost item and a labelled basket/traffic spillover on the uncovered share. Joybuy launched in March 2026 as 1P only, with a curated marketplace test described as planned from H2 2026.

2. Already-built warehouses are a sunk cost. Joybuy has opened warehouses at Milton Keynes and Luton. Their fixed cost is incurred whether a given SKU is stocked or not. The unit layer uses variable fulfilment only (pick, pack, last-mile). Under-utilisation is a scale-up investment, shown separately. Same-day subsidy is a customer-acquisition cost, also shown separately.

3. A land-grab layer sits on top of, and is separate from, unit economics. Even when unit contribution is thin, an entrant may still range as 1P to complete the offer, put volume through already-open sites, and build traffic before ads can monetise 3P. That strategic verdict is reported next to the unit verdict, so you can see the 1P-led mix an entrant runs and the unit contribution it accepts. As seller depth and ad income grow, the strategic layer gives way to unit economics, and the mix can shift toward 3P.

As seller coverage and the advertiser base grow, two things change. 3P becomes a real alternative (sellers can actually supply the range), and the 3P take gets richer (published-like commission, plus material ads, plus a warehouse margin). 1P economics also usually improve toward fully loaded at-scale fulfilment and benchmark margins. Mid-tail and thinner-margin lines then tend to move to 3P, while the platform may still stock the high-velocity, high-control head itself. Amazon's disclosed 61% of worldwide paid units (Q4 2025) is what a mature *unit* mix can look like. This model does not aim at that 61%, and it does not confuse it with JD China's 78.2% 1P revenue share, which is not a Joybuy figure and is not a share of units.

The unit-economics verdict and the strategic verdict are reported separately, on purpose. Under the default entrant assumptions the core mid-band range is 1P-led on both layers, because the real alternative to 1P is ‘not ranged’ and variable-only fulfilment with a 5% haircut leaves a thin but positive contribution. That is consistent with Joybuy’s 1P-only launch, and the model is not fitted to it. JD China’s 78.2% 1P revenue share is not Joybuy’s mix.

What this run is saying

Under scenario A (A · Entrant (Joybuy-like)), at the mid-price band (20 sub-categories), the operating (strategic) recommendations are Bring in-house 20, Test as 3P first 0, Keep 3P 0 (of 20). That is 100% of sub-categories, and 100% of modelled mid-band sales, on the 1P path (Bring in-house) — the mix an entrant would run. The unit-economics verdict on the same mid-band is Bring in-house 20, Test as 3P first 0, Keep 3P 0 (of 20), or 100% of sub-categories and 100% of

modelled sales on the 1P path. Where the two differ, the platform is accepting a thin (or negative) unit contribution for range completeness. Modelled sales are an Assumption (ex-VAT price times assumed weekly units), not observed GMV.

Under scenario B (B · Established (Amazon UK-like)), seller coverage is deep enough that the strategic layer follows unit economics, so the two verdicts coincide. Mid-band recommendations are Bring in-house 5, Test as 3P first 3, Keep 3P 12 (of 20). That is 25% of sub-categories and 6% of modelled mid-band sales on the 1P path. The sales-weighted 1P share can be lower than the sub-category share when the 1P recommendations sit in lower-priced lines (beauty, baby care) and the higher-priced durables stay on the 3P path.

15 of 20 mid-band sub-categories change operating recommendation between A and B: Laundry (washing machines and dryers) (Bring in-house under A, Keep 3P under B), Refrigeration (Bring in-house under A, Keep 3P under B), Cooking (ovens, hobs, cookers) (Bring in-house under A, Keep 3P under B), Compact appliances (kettles, microwaves, vacuums) (Bring in-house under A, Keep 3P under B), Laptops (Bring in-house under A, Keep 3P under B), Tablets (Bring in-house under A, Keep 3P under B), Peripherals (keyboards, mice, webcams) (Bring in-house under A, Keep 3P under B), Office equipment (printers, ink, desks) (Bring in-house under A, Keep 3P under B), Everyday clothing (Bring in-house under A, Keep 3P under B), Outerwear (Bring in-house under A, Test as 3P first under B), Footwear (Bring in-house under A, Test as 3P first under B), Sportswear (Bring in-house under A, Keep 3P under B), Baby clothing (Bring in-house under A, Test as 3P first under B), Pushchairs (Bring in-house under A, Keep 3P under B), Car seats and safety (Bring in-house under A, Keep 3P under B).

Sanity check, not a fitted target. Under the default entrant assumptions the operating mix on the mid-band core range is 1P-led (20 of 20 Bring in-house, 100% of sub-categories and 100% of modelled sales). The unit-economics verdict on the core mid-band range agrees (20 of 20 Bring in-house). 0 of 20 mid-band sub-categories have a negative 1P contribution (minimum 5.6%, median 12.5% of the ex-VAT price). Electronics sit at a thin but positive contribution, around six percent, because fulfilment in the unit figure is variable-only and the margin haircut is 5%, not 25%. This is consistent with Joybuy's observed 1P-only launch (March 2026), and the model was not fitted to that launch. The result is driven by treating 3P as unavailable where seller coverage is zero (the real alternative is 'not ranged'), by a 5% relative margin haircut justified by JD's published supplier-scale statements, and by keeping already-built warehouse fixed cost out of the per-SKU unit figure. JD China's 78.2% 1P revenue share is a parent-company figure, not Joybuy's mix, and is not used as a target.

Beauty clears the 1P money test in both scenarios under these assumptions: THG Beauty's 39.3% adjusted gross margin, even after THG's 12.3% group distribution proxy and the small entrant haircut, leaves more contribution than either 'not ranged' (at the entrant, where coverage is zero) or a mature 3P take (Amazon UK's 15% beauty referral on prices above £10, grossed up for VAT, plus a material ad take). Baby care, which uses the same THG proxy, follows the same pattern on the established side and is 1P at the entrant too.

Electronics and appliances are the clearest illustration of the two layers. At the entrant, Currys' 18.3% group gross margin with a 5% haircut, variable-only fulfilment (70% of the 12% established estimate) and a zero seller-coverage alternative ('not ranged') leaves a thin but positive 1P contribution of around six percent of selling price, so both the unit and the strategic verdict are Bring in-house. Once the platform is established, fulfilment is fully loaded, seller coverage is high, and a mature 3P take (Amazon's 7% referral grossed up for VAT to 8.4% of ex-VAT, plus ads and a warehouse margin) beats that thin 1P contribution, so the verdict flips to Keep 3P. Currys' 18.3% is a group figure including services and the Nordics, and the 12% fulfilment rate is a model estimate: both are open questions. The model does not invent a better electronics margin.

Clothing is ranged as 1P at the entrant (Zalando's 39.9% gross margin, a 5% haircut, and variable-only fulfilment leave a mid-teens contribution that beats 'not ranged'). Once established, Zalando's 23.0% fulfilment-cost ratio (an EU figure, already reflecting fashion returns) uses most of the margin, and a mature clothing take (Amazon UK's tiered referral, plus ads) is also high, so the verdict moves to Keep 3P or Test as 3P first. This is consistent with fashion being a natural later 3P expansion.

As platform maturity rises, seller coverage, advertising take and commission grow, and 1P fulfilment moves from variable-only toward a fully loaded at-scale rate. The maturity slider shows the operating mix starting 1P-led (a land-grab with thin unit contribution on durables) and moving toward more 3P as seller depth and ad income grow. Beauty tends to stay 1P; durables and fashion flip toward 3P once 3P is a real alternative. Amazon has disclosed that third-party sellers accounted for 61% of worldwide paid units in Q4 2025; this model is not fitted to that 61%, and its modelled-sales mix is an assumption-driven output, not a forecast of Joybuy's mix. JD China's 78.2% 1P revenue share is likewise a parent figure, not a Joybuy target.

Nuance

Contribution is not profit. The 1P unit figure is selling price minus cost of goods, variable fulfilment (at the entrant) or fully loaded fulfilment (once established), a CV-scaled extra markdown and physical holding. It does not subtract central overhead, marketing, the same-day subsidy, unallocated warehouse fixed cost, or tax. The 3P figure is what the platform

keeps (commission, plus ads, plus a warehouse margin) where sellers can actually supply, not what the seller pays.

3P is only a real alternative where seller coverage is material. When coverage is near zero, the unit test compares 1P (contribution plus a labelled basket/traffic spillover) with 'not ranged' (payoff zero). It does not compare 1P with a 3P take that is not available. The spillover default is 8% of the ex-VAT price, an Assumption.

Already-built warehouse fixed cost is sunk. The unallocated fixed-cost memo on the entrant mid-band is around 3.6% of the ex-VAT price (a scale-up investment, not a per-SKU penalty). The same-day subsidy is around 2.0% of price and is a customer-acquisition cost, also shown separately and not inside the unit verdict.

Return on stock is the second money test. A positive 1P contribution can still fail if inventory turns too slowly. The 25% hurdle is an Assumption, set above a typical UK retailer's cost of capital (usually high single digits) because it is a return-on-inventory bar, not a WACC. Currys' published 60 stock days are the electronics cover; fashion and beauty use an assumed 50 days. Joybuy has not published a hurdle or a stock-days figure.

Demand risk is a cost and a warning, not an automatic override. Assumed weekly CV raises safety-stock days and an extra markdown. The mid-tail rule (Jiang, Jerath and Srinivasan, 2011) leaves unproven or long-tail lines unranged when there are no sellers to learn from, or sends them to 3P first when coverage is material. You can turn that rule off.

The unit-economics verdict and the entrant strategic verdict are reported separately. The operating mix — what an entrant would run — uses the strategic layer while seller coverage is thin. That layer is for range completeness, warehouse fill, and the absence of 3P ad monetisation. It is not folded into the unit contribution.

Stocking a line as 1P can cannibalise 3P sellers of the same goods, and opening a line to 3P can cannibalise 1P. The model compares contribution on a single representative unit. It does not model competitive response, buy-box rules, or a seller's decision to leave the platform.

At the entrant end, mid-band 1P contribution ranges from 5.6% to 26.1% of the ex-VAT price. At the established end it ranges from 2.9% to 24.4%. A thin positive contribution on durables is a legitimate entrant outcome under these Assumptions, and it is labelled as such.

Platform maturity and the mix shift

The optional platform-maturity slider in the app interpolates linearly from A to B: seller coverage rises, the 1P margin haircut fades, fulfilment moves from variable-only toward a fully loaded at-scale rate, commission blends toward Amazon UK published referral, and advertising and the warehouse margin move toward their established values. The chart below is the implied 1P mix along that path — operating (strategic) and unit-economics — under the default assumptions. The mix starts 1P-led and moves toward more 3P as seller depth and ad income grow.

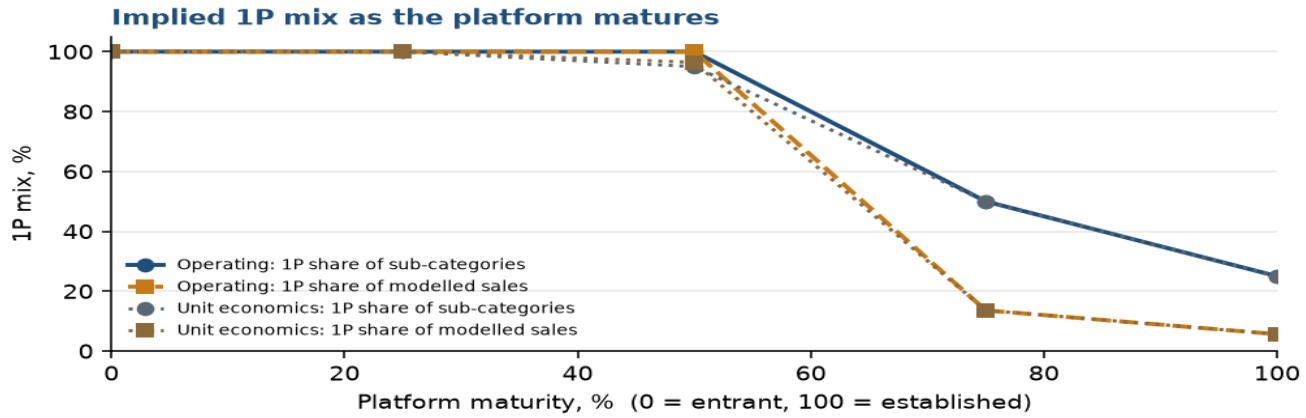


Figure 2. Implied 1P mix along the platform-maturity slider. Solid and short-dash lines are the operating (strategic) mix — what an entrant would run. Dotted lines are the unit-economics mix. The two converge as seller coverage crosses the 50% Assumption threshold. The sales-weighted share can sit well below the sub-category share when 1P recommendations are in lower-priced categories. All series are model outputs on assumed prices and weekly units, not observed GMV, and not fitted to Amazon's disclosed 61% third-party share of paid units or to JD China's 78.2% 1P revenue share.

Decisions per sub-category, under each scenario

One mid-price band per sub-category. Prices are Assumptions in GBP, ex-VAT. Contribution is a share of that ex-VAT price. Each scenario has a unit-economics verdict and a strategic verdict. The operating recommendation is the strategic one while seller coverage is thin. A flip means the operating recommendation changes between A and B.

Category	Sub-category	A 1P contrib.	A unit	A strategic	B 1P contrib.	B unit	B strategic	Flip
Home appliances	Laundry (washing machines and dryers)	6.2%	Bring in-house	Bring in-house	3.5%	Keep 3P	Keep 3P	Yes
Home appliances	Refrigeration	6.2%	Bring in-house	Bring in-house	3.5%	Keep 3P	Keep 3P	Yes
Home appliances	Cooking (ovens, hobs, cookers)	6.0%	Bring in-house	Bring in-house	3.3%	Keep 3P	Keep 3P	Yes
Home appliances	Compact appliances (kettles, microwaves, vacuums)	6.0%	Bring in-house	Bring in-house	3.3%	Keep 3P	Keep 3P	Yes
Computers and office	Laptops	6.2%	Bring in-house	Bring in-house	3.5%	Keep 3P	Keep 3P	Yes
Computers and office	Tablets	6.2%	Bring in-house	Bring in-house	3.5%	Keep 3P	Keep 3P	Yes
Computers and office	Peripherals (keyboards, mice, webcams)	5.8%	Bring in-house	Bring in-house	3.1%	Keep 3P	Keep 3P	Yes
Computers and office	Office equipment (printers, ink, desks)	5.8%	Bring in-house	Bring in-house	3.1%	Keep 3P	Keep 3P	Yes
Beauty	Skincare	26.1%	Bring in-house	Bring in-house	24.4%	Bring in-house	Bring in-house	—
Beauty	Makeup	25.9%	Bring in-house	Bring in-house	24.2%	Bring in-house	Bring in-house	—
Beauty	Haircare	26.1%	Bring in-house	Bring in-house	24.4%	Bring in-house	Bring in-house	—
Beauty	Fragrance	25.9%	Bring in-house	Bring in-house	24.2%	Bring in-house	Bring in-house	—
Clothing	Everyday clothing	18.8%	Bring in-house	Bring in-house	13.9%	Keep 3P	Keep 3P	Yes
Clothing	Outerwear	18.8%	Bring in-house	Bring in-house	13.9%	Test as 3P first	Test as 3P first	Yes
Clothing	Footwear	19.0%	Bring in-house	Bring in-house	14.1%	Test as 3P first	Test as 3P first	Yes
Clothing	Sportswear	19.0%	Bring in-house	Bring in-house	14.1%	Keep 3P	Keep 3P	Yes
Mother and baby	Baby care (toiletries, nappies, feeding)	26.1%	Bring in-house	Bring in-house	24.4%	Bring in-house	Bring in-house	—
Mother and baby	Baby clothing	19.0%	Bring in-house	Bring in-house	14.1%	Test as 3P first	Test as 3P first	Yes
Mother and baby	Pushchairs	5.6%	Bring in-house	Bring in-house	2.9%	Keep 3P	Keep 3P	Yes
Mother and baby	Car seats and safety	5.8%	Bring in-house	Bring in-house	3.1%	Keep 3P	Keep 3P	Yes

Table 1. 15 of 20 mid-band sub-categories change operating recommendation between the entrant and the established scenario (15 flip on the unit-economics verdict). Model estimates on assumed price points, not Joybuy decisions.

Implied 1P mix by category

Category	A: 1P subs	A: 1P share of subs	A: 1P share of sales	B: 1P subs	B: 1P share of subs	B: 1P share of sales
Home appliances	4 of 4	100%	100%	0 of 4	0%	0%
Computers and office	4 of 4	100%	100%	0 of 4	0%	0%
Beauty	4 of 4	100%	100%	4 of 4	100%	100%
Clothing	4 of 4	100%	100%	0 of 4	0%	0%
Mother and baby	4 of 4	100%	100%	1 of 4	25%	10%

Table 2. Operating (strategic) 1P path share of sub-categories and of modelled mid-band sales, by category. Under the default entrant assumptions the unit-economics mix on the mid-band agrees (20 of 20 Bring in-house). Modelled sales are an Assumption.

Breakeven chart

Each point is a mid-band sub-category. The vertical axis is 1P contribution as a share of the ex-VAT price (the 3P take at which the platform is indifferent on contribution). The horizontal axis is the effective 3P take. Circles are the entrant (A); squares are the established marketplace (B). Points above the dashed diagonal have 1P ahead on contribution. Contribution can be negative.

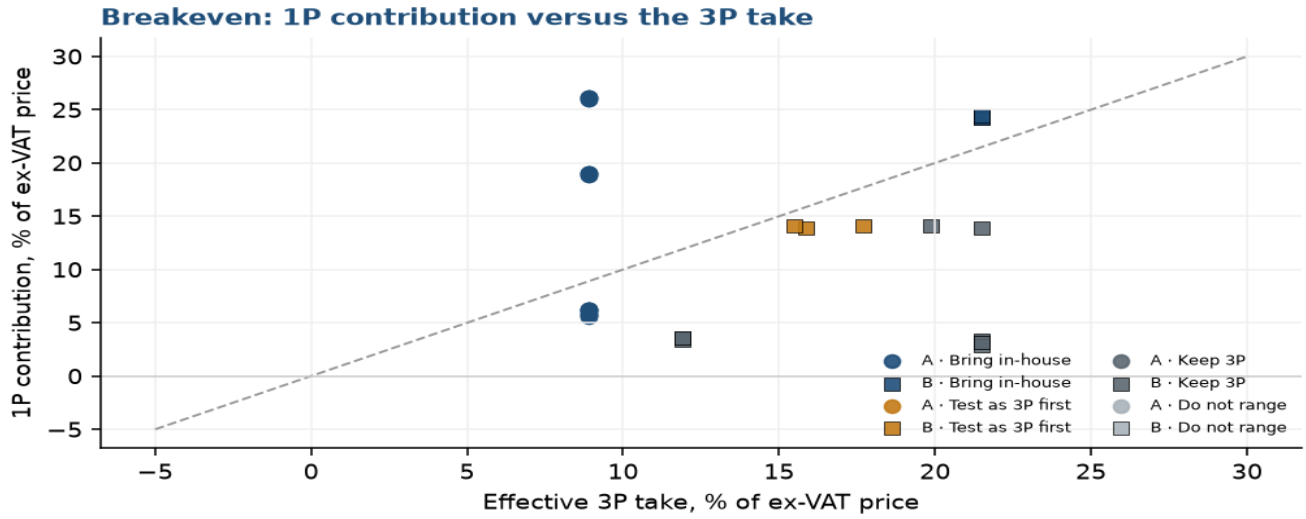


Figure 3. 1P contribution versus the 3P take, mid band, scenarios A and B.

The four tests, and the decision rules

1. Contribution per unit, availability-adjusted. What is kept if we stock it (plus a labelled basket/traffic spillover), versus the 3P take only where sellers can actually supply. Where they cannot, the alternative is ‘not ranged’ (payoff zero, plus a lost-assortment value), not a 3P take. 1P contribution is ex-VAT price, minus cost of goods at the (maturity-adjusted) gross margin, minus variable fulfilment at the entrant (or fully loaded fulfilment at scale), minus a CV-scaled extra markdown, minus physical holding. Already-built warehouse fixed cost and the same-day subsidy are memo lines, not in the unit verdict. The 3P take is commission (introductory Assumption at the entrant end; Amazon UK published referral as a benchmark stand-in at the established end, applied on the VAT-inclusive price) plus advertising and the platform’s warehouse margin, only for the assumed share of 3P sales fulfilled through Milton Keynes and Luton.

2. Inventory liquidity. Stock days and turn. Currys’ published 60 stock days (Published (Currys plc Annual Report & Accounts 2025/26), page 50) for electronics and durables; an assumed 50 days for fashion and beauty.

3. Return on stock (GMROI). (1P contribution × 52) / (cost of goods × effective cover in weeks), tested against a 25% hurdle. The 25% return-on-stock hurdle used in the money test is an Assumption, not a published WACC. A listed UK retailer's weighted average cost of capital typically sits in the high single digits (third-party estimates vary and are not used as a model input). The inventory hurdle is a return-on-stock bar, which a retailer may set well above WACC to cover risk, markdowns and scarce warehouse space. It is not equal to the cost of capital, and it is not Joybuy's published hurdle (Joybuy has not published one).

4. Demand predictability. Assumed weekly velocity and CV drive the mid-tail rule (Jiang, Jerath and Srinivasan, 2011) and a safety-stock add-on. Volatility is a cost and a warning, not an automatic override. Long-tail lines with no sellers are left as a hole. A land-grab layer, clearly separated from unit economics, ranges mid- and high-velocity lines as 1P while seller coverage is below 50%. A close-case strategic tilt (Hagiu and Wright, 2015) never overturns a clear money loss on the unit layer.

Compare 1P contribution with the 3P take

The paper compares agency selling (the supplier sells on the platform and pays a commission; the platform does not take inventory) with reselling (the platform buys and resells, sets the price, and earns a merchandise margin). Agency is more channel-efficient and tends to lower retail prices, but hands pricing control to the supplier. The platform's choice depends on which format yields more, given costs, control, and cross-channel effects. That is the basis for comparing 1P contribution with the 3P commission take. Abhishek, Jerath and Zhang (2016), Management Science 62(8), 2259–2280.

In this model. 1P contribution is the ex-VAT selling price, minus cost of goods at the (maturity-adjusted) gross margin, minus fulfilment, minus a CV-scaled extra markdown, minus physical holding. The 3P take is the commission layer (an assumed introductory rate at the

entrant end, or Amazon UK published referral as a labelled benchmark stand-in at the established end, applied on the VAT-inclusive price) plus, in the full-take run, a model-estimated advertising take and the platform's warehouse margin on an assumed share of sellers using Milton Keynes and Luton. Bring in-house needs 1P to beat that take. A fees-only mode drops ads and the warehouse margin.

Return on inventory has to clear a hurdle

Retailers judge 1P not only on unit contribution but on what the stock earns on the capital it ties up (GMROI, or return on inventory). Currys publishes group stock days of 60 (Annual Report 2025/26, page 50), used as the electronics cover. The 25% return-on-stock hurdle used in the money test is an Assumption, not a published WACC. A listed UK retailer's weighted average cost of capital typically sits in the high single digits (third-party estimates vary and are not used as a model input). The inventory hurdle is a return-on-stock bar, which a retailer may set well above WACC to cover risk, markdowns and scarce warehouse space. It is not equal to the cost of capital, and it is not Joybuy's published hurdle (Joybuy has not published one). Published (Currys plc Annual Report 2025/26), page 50, stock days 60.

In this model. Return on stock = (1P contribution × 52) / (cost of goods × effective cover in weeks). Effective cover is the category base (Currys 60 days for electronics and durables; an assumed 50 days for fashion and beauty) plus safety stock that scales with assumed CV.

High-volume, predictable lines to 1P; the unproven tail to 3P first

The paper records that Amazon tends to stock and sell high-demand products itself and leave long-tail products to independent sellers who pay a fee. For mid-tail products, where demand is uncertain in advance, the platform may learn from early third-party sales which items are worth selling directly. That is the basis for the model's 'test as 3P first on the unproven tail' rule: it is a learning step, not an automatic rejection of 1P. Jiang, Jerath and Srinivasan (2011), Marketing Science 30(5), 757–775.

In this model. Assumed weekly velocity is labelled high, mid or low, and is an assumption, not observed sales. If 1P wins the money tests but assumed demand is a long tail (low velocity), the recommendation is Keep 3P. If assumed demand is mid-velocity with a high weekly swing, the recommendation is test as 3P first. High-velocity, money-winning lines go to 1P. You can turn this rule off.

Category control, only in close cases

The paper treats the 1P vs 3P choice as a question of who should hold control rights over non-contractible decisions (marketing, pricing, fulfilment, service): the intermediary (reseller / 1P) or the supplier (marketplace / 3P). Reseller mode is preferred when the intermediary has the more important information, or when activities spill over across products. Marketplace mode is preferred when independent suppliers have better product-specific information (for example, style and fit in fashion). That is the basis for a documented strategic tilt in close cases: electronics and appliances, where JD's fulfilment and authenticity are named strengths, lean 1P; fashion, where suppliers often know the SKU better, leans 3P. Hagiu and Wright (2015), Management Science 61(1), 184–203; Joybuy launched as 1P-only in March 2026 (Operator One, September 2026; ChannelX, June 2026).

In this model. A documented tilt applies only when a money test is already within 10% of the line. Electronics, appliances, pushchairs and car seats tilt to 1P (control, safety, warehouse fill). Fashion stays as test-first (sellers often have better style and fit information). Beauty and baby care, where authenticity matters, tilt to 1P if demand is not a long tail. The tilt never overturns a clear money loss, and you can turn it off.

Volatility is a cost and a warning, not an automatic override

A weekly coefficient of variation above 1 means the standard deviation of weekly units is larger than the average itself: demand is, in that sense, highly variable. This is a conventional statistical bar, not a figure from a filing. The model uses it as a warning only. Extra safety stock and an extra markdown already scale with each archetype's assumed CV, so more variable lines are treated more cautiously in the money tests whether or not they cross the bar. In this version, weekly CVs are assumptions, not observed sales. The app also reports decisions if the bar is 0.75 or 1.5. Standard safety-stock rule: extra units = z × demand standard deviation over the lead time, which in weekly terms is z × CV × average weekly units × sqrt(lead time in weeks). Expressed in days of cover: z × CV × sqrt(L) × 7. z = 1.65 corresponds to a 95% cycle-service level under a normal-demand assumption. Lead time of 2 weeks is an assumption. This is an operations formula, not a figure from a company filing. Operations formula (safety stock) and a conventional CV > 1 warning bar; not a filing figure.

In this model. Assumed weekly CV raises safety-stock days and an extra markdown of 4% × CV (an assumption). A warning is raised when assumed CV is above 1. The warning does not, by default, change the recommendation. In this version CVs are assumptions, not observed series.

Entrant versus established marketplace

A new marketplace and a mature one do not have the same 1P costs or the same 3P take. Joybuy launched 1P-only; Amazon has disclosed that third-party sellers accounted for 61% of worldwide paid units in Q4 2025. Joybuy launch: Operator One, September 2026; ChannelX, 15 June 2026. Amazon seller unit mix: Q4 2025 earnings release, 5 February 2026.

In this model. Scenario A (entrant) uses a small margin haircut, variable-only fulfilment (sunk warehouse fixed costs shown separately), a near-zero ad take, an introductory commission, and — the central v3.1 fix — seller coverage that is near zero, so the real alternative to 1P is 'not ranged'. Scenario B uses published retailer benchmarks, fully loaded fulfilment, Amazon UK published referral as a stand-in, a material ad take, and high seller coverage. A maturity slider interpolates between them. A land-grab layer, shown separately from unit economics, ranges the core offer as 1P while coverage is thin. All entrant numbers are Assumptions with stated reasoning.

3P is only a real alternative where sellers can supply the range

A 3P take is not the opportunity cost of 1P if there is no seller to take the listing. Joybuy launched 1P-only in March 2026; a curated marketplace test was planned from H2 2026. At that point the alternative to stocking a line is a hole in the range, which also risks a lost basket. Joybuy launch: Operator One, September 2026; ChannelX, 15 June 2026. The coverage ratios themselves are Assumptions.

In this model. Seller coverage is the assumed share of the sub-category range that third-party sellers can actually supply (default 0% at the entrant end, 90% at the established end). The unit-economics 3P payoff is coverage × (3P take + spillover). The uncovered share is valued as 'not ranged': the platform gets no take and loses a labelled basket/traffic spillover (default 8% of the ex-VAT price). 1P payoff is 1P contribution plus the same spillover (the item is ranged). This is unit economics done with the right alternative, not a strategic fudge.

Already-built warehouse fixed cost is sunk

Joybuy has opened warehouses at Milton Keynes and Luton. Their occupancy and base-staff cost is incurred whether a given SKU is stocked or not. The incremental 1P decision should use variable fulfilment (pick, pack, last-mile). Under-utilisation is a network-level scale-up investment. JD's 20-F discusses improving overall utilisation through economies of scale as a network goal, not a SKU penalty. Warehouses: ChannelX, June 2026; Operator One, September 2026. JD 20-F FY2025, Item 5, utilisation through economies of scale. The 70/30 variable/fixed split is an Assumption.

In this model. Entrant unit contribution uses 70% of the established fulfilment rate. The remaining 30%, and any extra from under-utilisation, is reported as a scale-up investment memo, not subtracted from the per-SKU contribution. A same-day subsidy is a customer-acquisition cost, also shown separately and not in the unit verdict. At the established end, fulfilment is fully loaded at scale.

A land-grab layer, separate from unit economics

An entrant may range as 1P to complete the offer, fill already-open warehouses, and build traffic before ads can monetise 3P — even when unit contribution is thin. That is a strategic choice, and it is reported as such. Joybuy launched as 1P-only in March 2026 (Operator One; ChannelX). That launch is published context for a sanity check, not a target this model is fitted to. JD China's 78.2% 1P revenue share is a parent-company figure, not Joybuy's mix.

In this model. When seller coverage is below 50% (an Assumption), the strategic verdict for mid- and high-velocity lines is Range as 1P (Bring in-house). Low-velocity long-tail lines may be left unranged. When coverage is at or above 50%, the strategic verdict follows unit economics. Both verdicts are shown. The operating mix — what an entrant would run — uses the strategic verdict.

Data sources

Every published figure has a document and a location. Joybuy seller commission is not among them.

Joybuy launch and unpublished fees

Joybuy launched in March 2026 in the UK, Germany, the Netherlands, France, Belgium and Luxembourg as a first-party (1P) retail offer. A curated marketplace test was described as planned from the second half of 2026 (Joybuy spokesperson to The Grocer, 9 June 2026, as reported by Operator One). Fulfilment warehouses are at Milton Keynes and Luton. Sellers, when invited, may use Joybuy fulfilment or self-fulfil. Same-day delivery has been described as 'order by 11am' (ChannelX, June 2026). The UK entity is Jingdong Retail (UK) Limited.

Joybuy has not published a seller commission or fee card. Operator One ('Sell on Joybuy: An Invited Partner's Guide', September 2026) states that Joybuy had not published fees, commission or seller requirements as of September 2026. ChannelX (June 2026) similarly reported that marketplace fees were not then available. The public Joybuy partner page (joybuy.co.uk/help/partner-with-Joybuy) does not list commission rates. Circulating 5–10% USD figures are unverified and are not used as published rates. Until Joybuy publishes its own card, the 3P commission in the established scenario is the Amazon UK published referral schedule, labelled as a benchmark stand-in.

VAT

Assumed selling prices in this model are exclusive of VAT, in line with how UK company accounts report revenue. Amazon UK referral fees are charged as a percentage of the total sales price, which is VAT-inclusive. The model therefore applies the published Amazon referral schedule to the VAT-inclusive equivalent (ex-VAT price × 1.20) and reports the effective take as a share of the ex-VAT price. A 7% referral on a VAT-inclusive price is 8.4% of the ex-VAT price.

Figure	Value	Where	Label
Currys group gross margin	18.3% (1,697 / 9,254)	Annual Report 2025/26, Note 3, page 132 (period ended 2 May 2026). Group figure, including services and the Nordics. UK&I EBIT 2.9% is not used as GM.	Published (Currys plc Annual Report & Accounts 2025/26)
Currys stock days	60 days (from 62)	Annual Report 2025/26, Performance review, page 50.	Published (Currys plc Annual Report & Accounts 2025/26)
THG Beauty adjusted GM	39.3% (FY2025; 40.0% in FY2024)	THG FY2025 results: 'Adjusted gross profit margin for the year stood at 39.3% (2024: 40.0%)'. Multi-geography.	Published (THG FY2025 results)
THG group distribution	12.3% of group revenue (from 13.1%)	THG FY2025 results. Group figure (Beauty and Nutrition), used as a beauty fulfilment proxy.	Published (THG FY2025 results)
Zalando gross margin	39.9% (40.7% in 2024); cost of sales 60.1% of revenue	FY2025 combined management report, page 82. EU figure.	Published (Zalando SE Annual Report 2025)
Zalando fulfilment	23.0% of revenue (22.9% in 2024)	FY2025 combined management report, page 82. EU figure.	Published (Zalando SE Annual Report 2025)
Amazon WW seller unit mix	61% of worldwide paid units (Q4 2025)	Q4 2025 earnings release, 5 February 2026. Excludes Whole Foods. The FY2025 10-K does not repeat this metric.	Published (Amazon Q4 2025 earnings release, 5 February 2026)
Amazon advertising	US$68,635 million (FY2025)	Amazon 10-K FY2025. Sold to sellers, vendors, publishers, authors and others. 3P GMV is not published.	Published (Amazon 10-K FY2025)
Amazon UK referral fees	7% to 15% by category, with clothing and beauty tiers; minimum £0.25	sell.amazon.co.uk/pricing. Used as a labelled stand-in, not as Joybuy's fees. Applied on the VAT-inclusive price.	Benchmark: Amazon UK published fees
JD supplier / manufacturer scale	Qualitative (not a model input number)	JD 20-F FY2025, Item 4.B, page 88 (supplier, brand and partner relationships) and page 95 (direct manufacturer relationships; better procurement terms). Supports a small default entrant haircut. Joybuy is not named in the 20-F.	Published (parent company)
JD 1P product revenue share	78.2% of JD net revenues	JD 20-F FY2025. Parent-company China revenue share (1P gross, 3P net), clearly separated. Not Joybuy's mix, and not a target this model is fitted to.	Published (parent company; not a Joybuy figure)

Table 3. Published and benchmark figures used in the UK model.

Parent company context (JD.com in China) — not Joybuy UK figures

JD.com's FY2025 20-F reports net product revenues of RMB 1,023.8 billion (78.2% of total net revenues) and net service revenues of RMB 285.3 billion (21.8%). Product revenue is 1P, recorded gross. Service revenue includes marketplace commissions, marketing, logistics and other services, recorded net. 78.2% is therefore a share of *revenue*, not a share of units or of goods sold, and overstates 1P's share of goods because 3P GMV is not in the denominator. The 20-F does not publish GMV. These figures are parent company context for JD in China; they are not Joybuy UK operating numbers.

Entrant assumptions, with reasoning

Every numerical parameter on the entrant side is an Assumption. Please challenge them; they are the main open questions.

Parameter	Default	Reasoning
Seller coverage, entrant / established	0% / 90%	Assumption, default 0% at the entrant end, 90% at the established end. Joybuy launched in March 2026 as 1P only; a curated marketplace test was described as planned from the second half of 2026 (Joybuy spokesperson to The Grocer, 9 June 2026, via Operator One). At a launch-like point, third-party sellers cannot actually supply the range, so the real alternative to 1P is 'not ranged' (a lost item and, with a spillover assumption, a lost basket), not a 3P take. 0% is consistent with that published 1P-only launch; it is still an Assumption for any later invited-seller sliver (try 5–15% in the sensitivity). The established 90% is also an Assumption: a deep seller base, in the spirit of Amazon UK, not a published Amazon 'share of range that 3P can supply'. Amazon's 61% of worldwide paid units (Q4 2025) is a unit-mix figure, not this coverage ratio, and is not used as the coverage input.
Basket / traffic spillover	8% of ex-VAT price	Assumption, default 8% of the ex-VAT selling price. If a line is not ranged, a customer who came for it may leave and take the rest of the basket, and the site is a thinner destination. 8% is a round, reviewable value for that extra contribution, not a filing figure and not observed Joybuy or Amazon basket data. It is labelled and is applied only to the share of the range that would otherwise be missing (1 − seller coverage). When a 3P seller can supply the item, 3P also retains the basket, so the spillover is in both payoffs. Sensitivity 0–20%.
1P gross-margin haircut (relative)	5% (0–10% band; 25% stress test)	Assumption, default 5%, in a 0–10% band, with a 25% sensitivity. Joybuy is Jingdong Retail (UK) Limited, the European retail arm of JD.com. JD's FY2025 20-F (Item 4.B, page 88) says: 'We have established strong relationships with numerous suppliers, brands and partners. We leverage these relationships and our retail technology capability…' and (Item 4.B, page 95) 'Our ability to establish direct relationships with manufacturers enables us to provide high-quality products and obtain better procurement terms and access to hard-to-get products. Direct cooperation with manufacturers enables us to increase supply chain efficiency by minimizing supply chain costs.' A scaling UK site may still give a little away on promotional pricing and on local-assortment buying terms, so a small 5% relative haircut is the default. A 25% haircut, used in v3.0, is kept as a stress test; it is no longer the central case, because it treated Joybuy as a standalone buyer with no parent-scale supplier relationships.
Fulfilment variable-cost share	70% of the established rate, at the entrant	Assumption, default 70% variable (pick, pack, last-mile) and 30% fixed (warehouse occupancy, equipment, base staff), once a site is open. Neither Joybuy, Currys, THG nor Amazon publishes this split for UK fulfilment. 70/30 is a reviewable starting point. The entrant unit-contribution uses only the variable share of the established fulfilment rate. The 30% fixed share, and any under-utilisation of it, is shown as a scale-up investment line, not subtracted from the per-SKU contribution. At the established end the full rate is used, because at scale those fixed costs are part of ordinary per-unit economics. The v3.0 1.5× per-SKU fulfilment multiple has been retired: it charged under-utilisation to each SKU, which is the wrong treatment of a sunk site cost. JD's 20-F (Item 5) discusses 'improving overall utilization through economies of scale' as a network-level efficiency goal, which is how this model now treats it.
Entrant warehouse utilisation (memo)	50%	Assumption, default 50% of at-scale utilisation. Milton Keynes and Luton are already open, so their fixed cost is incurred whether a given SKU is stocked or not. The extra fixed cost per unit that would appear if we allocated under-utilisation to SKUs is reported as a scale-up investment memo: established fulfilment × fixed share × (1/utilisation − 1). It is not a per-SKU decision cost.
Same-day subsidy (CAC, memo)	2% of ex-VAT price	Assumption, default 2% of the ex-VAT selling price. Joybuy has offered same-day delivery (order by 11am, ChannelX, June 2026). A subsidy to win UK customers is a customer-acquisition cost, shown as a separate line, and is not subtracted from the unit contribution used for the unit-economics verdict. Joybuy has not published the cost of that promise.
Advertising take, entrant	0.3%	Assumption. An entrant with a thin seller base and little UK traffic has few advertisers. Sponsored-ads income is close to empty at launch; 0.3% of selling price is a small non-zero placeholder so the slider can move, not a published Joybuy ads rate.
Advertising take, established	3.0% (range 1.5–5%)	Model estimate, central 3.0%, range 1.5% to 5.0%. Amazon's FY2025 10-K reports advertising services of US$68,635 million (FY2025), sold to 'sellers, vendors, publishers, authors, and others' — not 3P-only. Amazon does not publish 3P GMV, so an official ads-to-GMV take cannot be computed and is not fabricated. 3% is a labelled central estimate for a mature marketplace with a deep advertiser base, used only in the established scenario and the mature end of the slider.
Introductory commission	7% (sensitivity 5–8%)	Assumption, default 7%, with a 5–8% sensitivity. Joybuy has not published a commission. A new marketplace may undercut a mature incumbent to attract sellers. 7% sits near the low end of Amazon UK's published referral schedule (7% on computers and full-size appliances, 15% on many other categories). This is not a Joybuy rate. At launch, seller coverage is so thin that this rate is often irrelevant to the real alternative, which is 'not ranged'.
Warehouse share, entrant / established	80% / 60%	Assumption. ChannelX reports that sellers may use Joybuy fulfilment or self-fulfil. An entrant that has already opened Milton Keynes and Luton has a reason to route a high share of whatever 3P volume exists through those sites. 80% is a high assumed share, not a published figure. At near-zero seller coverage this line is small, because there is little 3P volume to fulfil. The established share (60%) is also an assumption.

Parameter	Default	Reasoning
Logistics margin, entrant / established	3% / 10%	Assumption. The platform keeps only its margin on warehouse fulfilment, and only for the assumed share of 3P sales that go through Milton Keynes and Luton. A thin 3% margin is more plausible for an entrant than a mature 10%. At near-zero seller coverage this is a small number. The established 10% is also an assumption.

Table 4. Entrant (and some established) parameters. All Assumptions or model estimates, with reasoning.

Open questions for reviewers

This overview exists so that others can improve the model. The entrant assumptions are the most important things to challenge. Please send comments on any of the following.

1. Seller coverage at launch

The entrant default is 0% seller coverage, consistent with Joybuy's published 1P-only launch, and 90% at the established end. Both are Assumptions. Would you use a small invited-seller sliver (5–15%) even at the entrant end, and what established coverage would you use? Amazon's 61% of paid units is a unit mix, not this coverage ratio.

2. Lost-assortment spillover

The default basket/traffic spillover is 8% of the ex-VAT selling price, applied to the uncovered share of the range. It is an Assumption, not observed basket data. Is 8% too high, too low, or the wrong shape (a fixed basket, a multiple of item contribution)? What happens to the unit verdict at 0% and at 20%?

3. Entrant gross-margin haircut

The default is now a 5% relative haircut (0–10% band), justified by JD's published supplier and manufacturer relationships, not the v3.0 25% (which is kept as a stress test). Is 5% fair for a UK scale-up that can lean on parent buying, or would you use 0% or 10%?

4. Variable / fixed fulfilment split

Unit contribution at the entrant uses 70% of the established fulfilment rate as a variable (pick, pack, last-mile) cost. The rest is treated as a sunk site cost and shown as a scale-up investment. Is 70/30 the right split? Would you put any of the under-utilisation back onto the SKU?

5. Same-day subsidy as customer-acquisition cost

A 2% of selling-price same-day subsidy is shown as a CAC line and is not in the unit verdict. Is that the right treatment, and is 2% the right order of magnitude?

6. The land-grab layer versus unit economics

When seller coverage is below 50%, the strategic verdict ranges mid- and high-velocity lines as 1P for completeness, warehouse fill and the absence of ads, even if unit contribution is thin. Is that the right threshold, and should long-tail lines be left as a hole? This layer is how the model can show a 1P-led operating mix without quietly changing the unit maths.

7. Currys 18.3% as an electronics GM

Currys' 18.3% is group gross margin (Note 3, page 132), including services, credit and the Nordics. The report does not publish a UK&I; or product-only GM. Is this an acceptable benchmark, or should electronics 1P use a lower product-only assumption?

8. Electronics fulfilment estimate and the variable share

Established online electronics fulfilment is a 12% model estimate. Currys does not publish an online fulfilment ratio; Amazon's 15.2% is fulfilment as a share of total net sales, including AWS. What rate, and what variable share, would you use?

9. Mother and baby proxies

No current public UK baby-specialist filing was used. Baby care borrows THG Beauty's 39.3%, baby clothing borrows Zalando's 39.9%, and pushchairs and car seats borrow Currys' 18.3%. Are those acceptable proxies?

10. Assumed prices and weekly demand

GBP price bands and weekly units are Assumptions, set in a plausible UK range, not observed Joybuy or Amazon sales, and not scraped from retailer listings. Do the mid-band prices look like a typical UK basket in each category?

11. Sanity check against Joybuy's 1P-only launch

Joybuy launched ~100% 1P. That is not JD China's 78.2% 1P revenue share, which is a parent-company figure (1P recorded gross, 3P net) and is not a Joybuy mix. The model is not fitted to 100%. If the entrant operating mix is 1P-led, which assumptions drive it (coverage, sunk fixed cost, haircut, spillover, the strategic layer)? If it is not, which assumption prevents it? Please challenge those, rather than asking the model to hide them.

Limits

Prices, weekly units and CVs are Assumptions, set in a plausible UK range, not observed Joybuy or Amazon sales, and not scraped from retailer listings. Joybuy has not published fees, 1P margins, fulfilment costs or a hurdle. Amazon UK fees are a labelled stand-in. Advertising takes are model estimates, because Amazon does not publish 3P GMV and sells ads to first-party vendors as well as to sellers. Currys' 18.3% is group gross margin, including services and the Nordics. Zalando figures are EU. THG Beauty is multi-geography. Established online electronics fulfilment (12%) is a model estimate. Mother and baby uses adjacent proxies (THG, Zalando, Currys). Seller coverage, the 8% spillover, the 70/30 variable/fixed fulfilment split, the 5% margin haircut and the 25% inventory hurdle are Assumptions. The model compares contribution on a single representative unit; it does not model competitive response, buy-box rules, or a seller's decision to leave. Amazon's disclosed 61% third-party share of worldwide paid units (Q4 2025) is published context for a mature mix; this model is not fitted to it. JD China's 78.2% 1P revenue share is a parent figure, not Joybuy's mix, and is not a target.

The interactive model (Streamlit) lets you move the platform-maturity slider, change seller coverage, the margin haircut, spillover, the variable-fulfilment share, introductory commission and advertising takes, turn the strategic layer off, and inspect a worked example for any sub-category. Run streamlit run streamlit_app.py from the repository.

Independent portfolio project on Joybuy's UK and European business. It uses published UK and EU retailer filings, Amazon UK published referral fees as a labelled benchmark stand-in (Joybuy has not published a seller fee card), stated Assumptions for the entrant scenario, and model-estimated advertising and warehouse-margin takes. This is decision support, not a forecast, and not a description of Joybuy's actual numbers or of any live SKU. Portfolio project. Made by Ami Sata.